

Firewall Policy

Ver. 4.0

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1. Purpose

Innova Solutions operates perimeter firewalls between the Internet and its private internal network to create a secure operating environment. A firewall is just one element of a layered approach to network security. The purpose of this Firewall Policy is to describe how firewalls will filter Internet traffic to mitigate risks and losses associated with security threats, while maintaining appropriate levels of access for business users.

2. Scope

The role of this firewall policy is to protect internal systems and restrict unwanted access into the Network. The firewall will (at minimum) perform the following security services:

- Access control between the trusted internal network and untrusted external networks.
- Block unwanted traffic as determined by the firewall rule set.
- Hide vulnerable internal systems from the Internet.
- Hide information, such as system names, network topologies, and internal user IDs, from the Internet.
- Log traffic to and from the internal network.
- Provide robust authentication.

3. Acronyms

SMTP	-	Simple Mail Transfer Protocol
DNS	-	Domain Name System
IP	-	Internet Protocol
ITOps	-	Information Technology Operations
SSH	-	Secure Shell
ISM	-	Information Security Manager
HTTP	-	Hypertext Transfer Protocol
VPN	-	Virtual Private Network

4. Policy

All critical applications at Innova Solutions Inc. shall be protected by firewall from both external users and internal users. The firewall shall be configured and managed to limit access to data only to authorized users.

4.1 Implementation Guidelines

1. Firewall Owner

1.1. ITOps Infrastructure manager shall be an assigned owner for the firewall.

2. Security of Firewall

2.1. The operating system and the firewall application shall be secured as per the respective Secure Configuration Documents.

2.2. The firewall system shall not be running any additional services that can be accessed remotely. Any additional service like SMTP, DNS would present attackers with an opportunity to compromise the firewall by exploiting vulnerabilities associated with these services.

3. Firewall Segments

3.1. Firewall shall segment the network based on the risk levels. Systems with similar risk level shall be put into one segment. For example, if the firewall is segregating the internal network from the Internet there shall be a minimum of three segments- one for Internet, one for internal network and one for machines that are accessed from both internal network and Internet.

3.2. Network Administrators along with the ITOps Manager and the Information Security Manager shall be responsible for determining the number of segments and the servers that will be located in each segment.

4. Firewall Rule Base Creation

4.1. Network Administrator shall be responsible for designing and testing the firewall rule-base before deployment.

4.2. Network Administrator shall get the required inputs from the respective project owner for designing the rule-base.

4.3. Firewall rule-base shall restrict access to required ports on the target machine. The source field in the rule-base shall be restricted to specific IP addresses/Subnet addresses wherever feasible. In the case of applications where the number of individual IP addresses/subnets is very large the source address can be made generic to make the rule base more manageable. Firewall shall have a rule to deny all access that is not explicitly allowed.

4.4. Access to administrative ports, including SSH and Microsoft Windows Terminal services on critical servers, shall be restricted to known source IP address.

4.5. User authentication provides additional security and shall be used for authenticating remote users for administrative system access.

4.6. Firewall user-database, needed for rules that are configured for user- authentication, if supported, can be stored either locally on the firewall or in an external directory server. Enforce password policy for these user accounts including password expiry, password history, and password complexity. Account lockout shall be configured to prevent password cracking attempts. It shall be ensured that these user credentials are transmitted in encrypted format from the user desktop to the firewall.

4.7. All rules shall be tested before being deployed. This would enable the firewall team to fine tune the rules related to the application.

4.8. Network Administrator shall give a copy of the tested rule-base to the project to the respective owners. This will ensure that project owners are aware of the services that will be allowed through the firewall prior to deployment. This will also help in reducing trouble shooting efforts when the firewall goes into production.

4.9. If there are applications that require large number of dynamic ports and access cannot be restricted to specific ports at the firewall, then these servers shall be separated out into a new segment.

5. Firewall Rule Base Change

5.1. After the firewall goes into production, all changes to the rule-base shall be made only after proper authorization by following the change management process, to ensure that the security level is maintained.

5.2. User shall contact the project owner for any access requirement. project owner shall validate the request, translate the user request to specific IP address and port numbers and pass it to the ITOps Manager.

5.3. The ITOps Manager shall analyze the request and make appropriate changes before approving the request. The Network Administrator shall be responsible for testing and implementing the changes and providing feedback to the concerned parties.

5.4. All changes to rules shall be tested before deployment. Changes can be tested either on a stand by firewall (if available) or on the production firewall itself by enabling more access than required and enabling logging for a limited time only.

5.5. The logs can be used to verify the exact port requirements. The activity of fine-tuning the ports based on Firewall logs shall be completed quickly as there is risk of unauthorized access during this time.

6. Administrative Access

6.1. Administrative access to Firewall application is required for activities, including:

6.1.1. Rule base modification

6.1.2. Firewall-user account management

6.1.3. Firewall-administrator account management, and

6.1.4. Log monitoring.

6.2. Access privileges shall be provided to assigned administrators on a need to have and need to do basis with explicit approval from ITOps Manager and Sr. Management.

6.3. Access to firewall administration programs shall be through encrypted channels. If the firewall itself does not provide this facility, then additional mechanisms like IP Sec shall be used for this purpose.

7. Logging

7.1. Logging shall be enabled to ensure that all critical access is tracked.

Logging shall be enabled for rules enabling administrative access (e.g., SSH access Intranet web server).

7.2. Logging shall be enabled for the last rule that blocks all access that is not explicitly allowed by the other rules. If there are known services like NetBIOS broadcast which generates large amount of log entries, a separate rule shall be added above the last rule to block this traffic without logging.

7.3. Logging shall be enabled to track any changes done to firewall configuration including changes to rule base. This will ensure that all changes can be tracked for trouble shooting as well as for audit purposes.

7.4. The IT Administrator shall monitor the logs periodically for:

- 7.4.1.** Port scans
- 7.4.2.** Authentication failures
- 7.4.3.** Denial of service attempts
- 7.4.4.** Failed connections

7.5. The IT Administrator shall give monthly reports on log findings to the ITOps Manager.

8. Performance Monitoring

8.1. Resource utilization shall be tracked to ensure that Firewall is performing at optimum level. Network Administrators shall monitor the critical system parameters on a continuous basis. Any surge in utilization of any of these parameters might be an indication of a system under attack. Network Administrator shall determine the threshold levels for peak and average usage for the following parameters:

- 8.1.1.** CPU utilization
- 8.1.2.** Memory utilization
- 8.1.3.** Hard disk free space
- 8.1.4.** Concurrent Connections

8.2. The Network Administrator shall give monthly reports on performance statistics to the IT Head/ ITOps Manager.

8.3. IDS/IPS Monitoring: Intrusion detection systems (IDSs) and Intrusion prevention system(IPSs) are the most sophisticated network security devices in use today. IDS/IPS inspect and prevent network packets and block suspicious ones, as well as alert administrators about attack attempts. These system logs contain valuable network threat information about attack types, devices being targeted, and more. ITOps Network Admin should monitor these logs and extract the information they provide to further secure the Innova Solutions Inc. network.

IDS/IPS security reports that provide information on:

- Attacks occurring on your network, with information on the most frequent attacks and the source of these attacks.
- The most targeted devices on your network.
- Attack trends.

ITOps Network admin reviews the alerts and performs a root cause analysis. ITOps Manager should review the root cause analysis and document the feedback.

9. Change Management

9.1. Changes to the following shall adhere to change management process:

- 9.1.1.** Installing a new patch on the firewall
- 9.1.2.** Installation or removal of additional component
- 9.1.3.** Integration of Firewall with third party components

9.1.4. Adding a new segment or modifying existing segment

9.1.5. Changes to the firewall rule base.

9.2. ITOps Manager shall be responsible for approving the changes. The Network Administrator shall be responsible for implementing the changes on the firewall.

10. Documentation

10.1. The Network Administrator shall maintain detailed documentation of the firewall architecture and administration tasks.

10.2. Firewall architecture documentation shall include the following:

10.2.1. Network diagram with firewall segments

10.2.2. IP addresses of firewall interface and network devices connected to Firewall

10.2.3. Routing table of Firewall and connected devices

10.2.4. Updated Rule-base

10.3. Documentation on firewall administration tasks, including the following:

10.3.1. Installation and configuration of firewall

10.3.2. Adding/Deleting/Modifying Rule base

10.3.3. Adding/Deleting/Modifying Routing table

10.3.4. Adding/Deleting/Modifying Firewall Users

10.3.5. Adding/Deleting/Modifying Network Administrators

10.3.6. Backup/Recovery of Firewall OS/Application files

10.3.7. Backup/Recovery of Firewall Rule-base

10.3.8. Backup/Recovery of log files

10.3.9. Backup/Recovery of user database

11. Backup and Redundancy

11.1. The Network Administrator is responsible for backup and recovery of Firewall.

The following shall be backed up soon after installation and successful testing of firewall:

11.1.1. Firewall OS files

11.1.2. Firewall application files

11.1.3. Configuration files

11.1.4. Firewall rule base

11.1.5. Routing Table

11.1.6. Firewall logs

11.2. The firewall logs shall be backed up daily.

11.3. Rule base shall be backed up before and soon after making any changes.

11.4. A fullback up of firewall application/operating system files shall be taken before any major changes to firewall, including:

11.4.1. Upgrade of Firewall OS/Application

11.4.2. Installation of any additional component on Firewall (e.g. VPN)

11.4.3. Integration of Firewall with third party components (e.g. Integration of Firewall with ITOps department for authentication)

11.4.4. Adding a new firewall interface

11.5. Firewall redundancy shall be maintained taking in to consideration, the criticality of the applications, services and networks being protected. If the need is critical, firewalls should be configured in high availability mode to ensure minimum down time for the respective applications.

11.6. If the firewall is not configured in high availability mode, recovery shall be tested once in six months on standby machine. It is not recommended to test recovery on live machine.

5. Policy Compliance

Compliance with this policy is mandatory for all employees of Innova Solutions Inc., and this policy is to be reviewed periodically. Compliance will be measured by the internal or external audits.

5.1 Exceptions

There are no exceptions to this policy.

5.2 Non-Compliance

Any employee, trainee, contractor, subcontractor, or committee found to be not complying with the requirements specified in the policy is bound to accept consequences specified in the Disciplinary Actions Specific Procedure.

6. Disclaimer

Company's Absolute Right to Alter or Abolish the Policy

Innova Solutions Inc. reserves the right in its absolute discretion to abolish the policy and dissolve it at any time, or to alter the terms and conditions. Such discretion may be exercised any time, before, during or after the policy year is completed.